Battery Health Monitoring & RUL Prediction for AUVs

Mentor Review 2

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Pre-Dataset Readiness & Literature Foundation

- Mapped marine AUV energy systems and subsea mission profiles (Ma et al., 2025)
- Ingested 34 NASA PCoE battery aging cells (2,744 cycles)
- Standardized feature extraction:
  - Voltage sag
  - Thermal rise (ΔT)
  - Energy delivered
  - Rest relaxation tags

Proposed Preprocessing, Modeling & Evaluation Plan

- CEEMDAN / Savitzky–Golay signal decomposition to decouple thermodynamic fade from regeneration rebounds
- 8-model architecture zoo:
  - Empirical, RF, LSTM, GRU, TCN
  - BiLSTM-Attention, Transformer, Hybrid SOTA
- Multi-task training engine with EarlyStopping and adaptive learning rate decay

Mentor Corrections, Validated Constraints & Next 3-Day Plan

- Dual threshold confirmed:
  - 1.40 Ah for B0005 / B0006 / B0018
  - 1.50 Ah for B0007
- Strict cell-wise split with frozen out-of-sample scaler fit
- Real-time AUV inference latency benchmark verified at < 1.6 ms